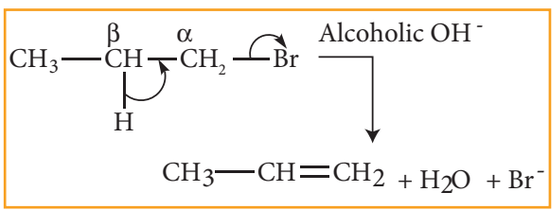



In the above reaction, Benzoyl peroxide acts as a radical initiator. The mechanism involves free radicals.

**Elimination reactions:**

In this reaction two substituents are eliminated from the molecule, and a new C=C double bond is formed between the carbon atoms to which the eliminated atoms/groups are previously attached. Elimination reaction is always accompanied with change in hybridisation.

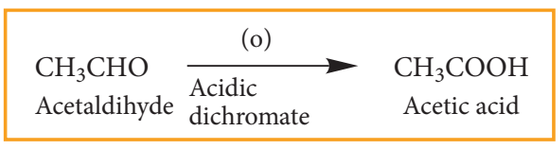
**Example:** n-Propyl bromide on reaction with alcoholic KOH gives propene. In this reaction hydrogen and Br are eliminated.

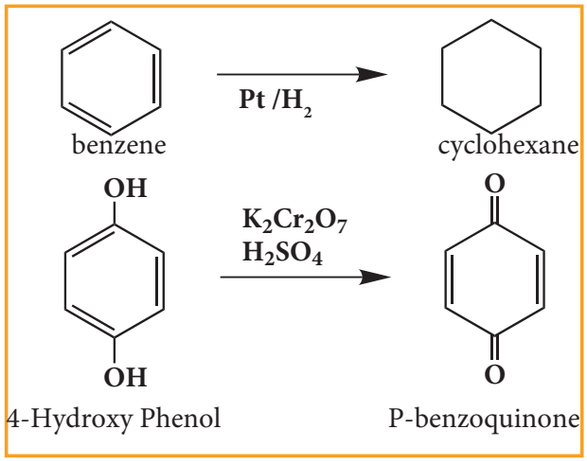



**Oxidation and reduction reactions:**

Many oxidation and reduction reactions of organic compounds fall into one of the four types of reaction that we already discussed but others do not. Most of the oxidation reaction of organic compounds involves gain of oxygen or loss of hydrogen Reduction involves gain of hydrogen and loss of oxygen.

**Examples:**







Apples contain an enzyme called polyphenol oxidase (PPO), also known as tyrosinase. Cutting an apple exposes its cells to the atmospheric oxygen and oxidizes the phenolic compounds present in apples. This is called the enzymatic browning that turns a cutapple brown. In addition to apples, enzymatic browning is also evident in bananas, pears, avocados and even potatoes

## 12.3 Functional Group inter conversion

Organic synthesis involves functional group inter conversions. A particular functional group can be converted into other functional group by reacting it with suitable reagents. For example: The carboxylic acid group (–COOH) presents in organic acids can be transformed to a variety of other functional group such as – $CH_2$–OH, – $CONH_2$, – COCl by treating the acid with $LiAlH_4$, $NH_3$ and $SOCl_2$ respectively.



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